Problem Set 4 — Mixed Models

Statistical Modelling (BAM), IE University — Spring 2027

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Instructions. Work in pairs; each student submits their own PDF (rendered from this `.Rmd`) on Blackboard. Show all derivations and interpret every number in context. Course notation: $y = X\beta + Zb + \varepsilon$, $b \sim N(0, G)$, $V = ZGZ^\top + \sigma^2 I$, ICC $= \sigma_b^2/(\sigma_b^2 + \sigma^2)$; fixed effects are *estimated*, random effects *predicted*. Packages: `lme4`, `nlme`, `MASS`; all data are in R packages. Expect 6–8 hours. Use `set.seed(2027)` wherever random numbers appear. Eight problems; each is worth the same.

1 Unequal groups: generalised least squares by hand

Four schools provide the following summary of a test score (the same test, pupils of the same age). Here n_i is the number of pupils sampled in school i and $\bar{y}_i$ their mean score.

School i	1	2	3	4
n_i	2	3	5	10
$\bar{y}_i$	52	60	55	63

Assume $y_{ij} = \mu + b_i + \varepsilon_{ij}$ with $b_i \sim N(0, \sigma_b^2)$, $\varepsilon_{ij} \sim N(0, \sigma^2)$ independent, and take $\sigma_b^2 = 16$, $\sigma^2 = 36$ as known.

(a) Show that $\mathbf{1}^\top V_i^{-1}\mathbf{1} = n_i/(\sigma^2 + n_i\sigma_b^2)$ and $\mathbf{1}^\top V_i^{-1} y_i = n_i\bar{y}_i/(\sigma^2 + n_i\sigma_b^2)$, where $V_i = \sigma^2 I_{n_i} + \sigma_b^2 \mathbf{1}\mathbf{1}^\top$. (Hint: $V_i\mathbf{1} = (\sigma^2 + n_i\sigma_b^2)\mathbf{1}$.)

(b) Deduce that the GLS estimator of μ is the weighted mean $\hat{\mu} = \sum_i w_i \bar{y}_i / \sum_i w_i$ with $w_i = n_i/(\sigma^2 + n_i\sigma_b^2) = 1/\mathrm{Var}(\bar{y}_i)$, and that $\mathrm{Var}(\hat{\mu}) = (\sum_i w_i)^{-1}$.

(c) Compute w_i, $\hat{\mu}$ and $\mathrm{se}(\hat{\mu})$. Compare with the ordinary mean of all 20 pupils, and compute the true variance of that ordinary mean under the model.

(d) What does $\hat{\mu}$ become when $\sigma_b^2 \to 0$ and when $\sigma_b^2 \to \infty$? Explain in words why the large school receives *less* than proportional weight.

(e) Verify your numbers in R by building the full 20×20 matrix V and computing $(X^\top V^{-1} X)^{-1} X^\top V^{-1} y$ with y equal to the school means repeated n_i times (this reproduces $\hat{\mu}$ because y enters only through the school means).

2 ML, REML and the ANOVA table: railway rails

The data `nlme::Rail` give the time (in nanoseconds, minus a constant) that an ultrasonic wave takes to travel along 6 randomly chosen steel rails, each measured 3 times (18 measurements). Model $y_{ij} = \mu + b_i + \varepsilon_{ij}$, $b_i \sim N(0, \sigma_b^2)$, $\varepsilon_{ij} \sim N(0, \sigma^2)$.

```
rail <- as.data.frame(nlme::Rail); rail$Rail <- factor(as.character(rail$Rail))
head(rail, 6)
```

```
#>   Rail travel
#> 1    1     55
#> 2    1     53
#> 3    1     54
#> 4    2     26
```

```
#> 5     2      37
#> 6     2      32
```

(a) **Derivation.** For a balanced layout with m groups of size n, let $\mathrm{MSW} = \mathrm{SSW}/\{m(n-1)\}$, $\mathrm{MSB} = \mathrm{SSB}/(m-1)$ and $\lambda = \sigma^2 + n\sigma_b^2$. Show that after an orthogonal transformation within each group the log-likelihood is, up to a constant, $-\frac{1}{2}\big[m(n-1)\log\sigma^2 + \mathrm{SSW}/\sigma^2 + m\log\lambda + \{\mathrm{SSB} + mn(\bar{y}-\mu)^2\}/\lambda\big]$, and derive the ML estimators of σ^2, λ and σ_b^2. Then explain which term must be removed to obtain the REML criterion and derive the REML estimator of σ_b^2.

(b) Compute the ANOVA table for the rail data (`aov`), and from it the REML and ML estimates of σ_b^2, σ^2 by the formulas of part (a). Check against `lmer(..., REML = TRUE/FALSE)`.

(c) Compute the ICC and a 95% profile-likelihood interval for σ_b and for σ. Interpret the variance components in nanoseconds.

(d) Give the estimate of μ and its standard error, and a 95% **prediction** interval for a *single new measurement on a new rail*, using $\mathrm{Var}(y_{\text{new}} - \hat{\mu}) = \mathrm{se}(\hat{\mu})^2 + \sigma_b^2 + \sigma^2$. Compute it in two ways: (i) the Wald (z) interval with ± 1.96; (ii) a t interval. In this balanced design $\mathrm{se}(\hat{\mu})^2 = \mathrm{MSB}/(mn)$ and $\sigma_b^2 + \sigma^2 = \mathrm{MSB}/n + \mathrm{MSW}(1 - 1/n)$, so the variance is a combination of two mean squares: use Satterthwaite degrees of freedom for it (for the *confidence* interval of μ the exact pivot is t_{m-1} with $\mathrm{se}^2 = \mathrm{MSB}/(mn)$; compare with `lmerTest`). (iii) Explain why the z interval is anti-conservative here.

(e) **Design.** With a budget of 18 measurements, compare $m \times n = 6 \times 3$, 9×2 and 3×6 for estimating μ using $\mathrm{Var}(\hat{\mu}) = (\sigma_b^2 + \sigma^2/n)/m$ with the REML estimates. What is the lesson?

3 BLUPs and shrinkage

(a) **Derivation.** In the random-intercept model $y_{ij} = \mu + b_i + \varepsilon_{ij}$ with $\mu, \sigma_b^2, \sigma^2$ known, show that $b_i \mid y_i \sim N\big(k_i(\bar{y}_i - \mu),\, k_i\sigma^2/n_i\big)$ with $k_i = n_i\sigma_b^2/(n_i\sigma_b^2 + \sigma^2)$. (Hint: $\bar{y}_i - \mu = b_i + \bar{\varepsilon}_i$ is sufficient for b_i; or use the conditional-normal formula with V_i^{-1} from Problem 1.)

(b) For the rail data (REML estimates from Problem 2, $\hat{\mu} = 66.5$) compute k_i, the BLUP $\tilde{b}_i$ and the conditional standard deviation for each rail. Verify with `ranef()`. Why are the BLUPs so close to the raw deviations $\bar{y}_i - \hat{\mu}$?

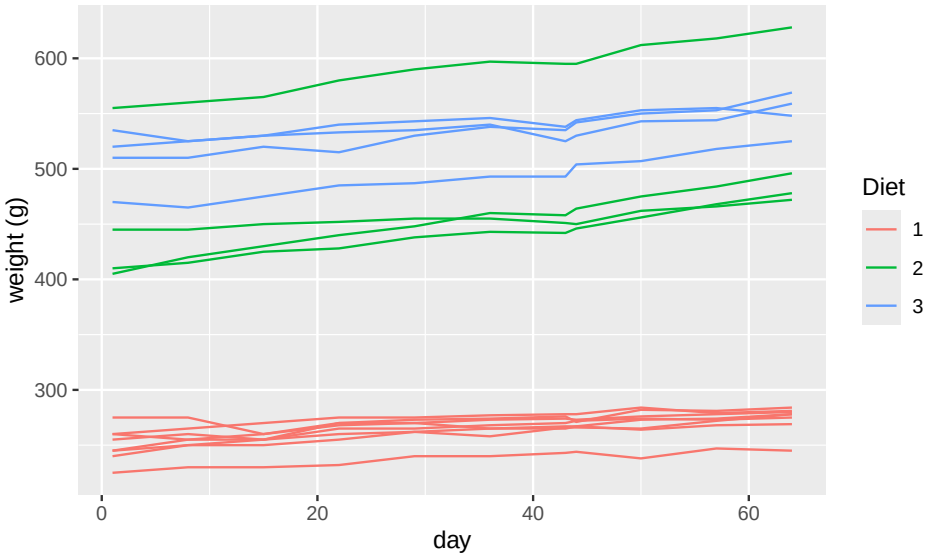
(c) Repeat the computation of k for `lme4::Dyestuff` (6 batches of 5; REML) and explain the difference with the rails.

(d) **Simulation.** With $m = 6$ groups, $n = 3$ observations each and the rail variance components, simulate 5000 data sets of group means $\bar{y}_i = b_i + \bar{\varepsilon}_i$ (take $\mu = 0$ known). Compare the mean squared error of the raw means $\bar{y}_i$ and of the BLUPs $k\bar{y}_i$ for predicting b_i. Repeat with the variance components and design of `Dyestuff` ($n = 5$). A colleague says "shrinkage introduces bias, so BLUPs are worse than raw means". Respond with your simulation.

4 Random slopes: growth of rats

The data `nlme::BodyWeight` give the body weight (g) of 16 rats measured at 11 times (days) over 64 days; the rats were fed one of three diets (8 on diet 1, 4 on diet 2, 4 on diet 3). Use the time variable `tc` = (`Time` − mean(`Time`))/10, i.e. time in tens of days, centred at the average observation time.

```
bw <- as.data.frame(nlme::BodyWeight); bw$Rat <- factor(as.character(bw$Rat)); bw$Diet
↪  <- factor(bw$Diet)
bw$tc <- (bw$Time - mean(bw$Time)) / 10
library(ggplot2)
ggplot(bw, aes(Time, weight, group = Rat, colour = Diet)) + geom_line() + labs(y =
↪  "weight (g)", x = "day")
```



(a) Fit `weight ~ tc * Diet + (tc | Rat)` by REML. Write the model in the notation of the course, report the fixed effects and all variance components, and interpret in grams and grams per ten days. Give the estimated growth rate (g per 10 days) for each diet with a 95% confidence interval for diet 1 only.

(b) Derive $\mathrm{Var}(y_{ij} \mid tc_{ij})$ as a function of tc. Compute its estimate at the first (Time $= 1$) and the last (Time $= 64$) day, compare with the random-intercept-only model, and comment.

(c) Test the need for the random slope (against a random intercept only, same fixed effects). State the null hypothesis, the reference distribution and the p-value, and say why χ^2_2 is not exactly right.

(d) Test the `tc:Diet` interaction with the appropriate likelihood. State the conclusion in words.

(e) The correlation $\hat\rho$ between random intercept and slope refers to `tc` $= 0$. Use the change-of-origin result to compute the standard deviation of the rat-specific *weight at Time = 1* and the correlation with the slope, and check by refitting with the origin at Time $= 1$.

(f) Predict, with a 95% prediction interval, the weight of a **new** rat on diet 2 at day 64. Give the Wald (z) interval and a t interval with Satterthwaite degrees of freedom (`lmerTest`), and say why the z interval is anti-conservative with 16 rats.

5 Fixed or random? Confounded group effects

Consider $y_{ij} = \beta x_{ij} + b_i + \varepsilon_{ij}$ with m groups of n observations, $\varepsilon_{ij} \sim N(0, \sigma^2)$, and suppose the group effect is related to the group mean of the covariate: $b_i = \gamma \bar{x}_i + u_i$, with $u_i \sim N(0, \sigma_u^2)$ independent of everything else.

(a) **Derivation.** Let $\hat\beta_W$ and $\hat\beta_B$ be the within and between estimators. Show that $\mathrm{E}[\hat\beta_W] = \beta$ and $\mathrm{E}[\hat\beta_B] = \beta + \gamma$ (conditionally on the x_{ij}). Using the weighted-average representation $\hat\beta_{GLS} = (S_W^{xx}\hat\beta_W + \theta S_B^{xx}\hat\beta_B)/(S_W^{xx} + \theta S_B^{xx})$ with $\theta = \sigma^2/(\sigma^2 + n\sigma_b^2)$, deduce the bias of the random-effects (GLS) estimator and of the pooled estimator ($\theta = 1$).

(b) **Simulation.** Take $m = 50$, $n = 4$, $\beta = 1$, $\sigma = 1$, $x_{ij} = z_i + w_{ij}$ with $z_i, w_{ij} \sim N(0,1)$ and $b_i = 0.8 z_i + u_i$, $u_i \sim N(0, 0.5^2)$. For 300 simulated data sets compute the slope estimates of (i) pooled `lm`, (ii) random intercept `lmer(y ~ x + (1 | id))`, (iii) fixed effects `lm(y ~ x + factor(id))` and (iv) `lmer(y ~ x + xbar + (1 | id))` where `xbar` is the group mean of `x`. Report mean and SD of each, and compare the bias of (i) and (ii) with the formula of part (a) (here $\gamma = \mathrm{Cov}(b, \bar x)/\mathrm{Var}(\bar x) = 0.8/1.25 = 0.64$; evaluate the formula with S_W^{xx}, S_B^{xx} and θ computed from each simulated data set, using the estimated variance components, and average).

(c) For **one** data set (the first of the simulation, seed `2027`) test whether the random-effects assumption is tenable by testing the coefficient of `xbar` in model (iv). What does the test compare? State the conclusion.

(d) Give two reasons why a researcher may still prefer a random-effects model with the group mean included over fixed effects, and one for which fixed effects are preferable.

6 Crossed random effects and a suspicious cell: factory workers

The data `nlme::Machines` contain the productivity score of 6 workers, each operating 3 machines (A, B, C) three times (54 observations). The three machines are of specific interest; the workers are a sample from the workforce.

```
ma <- as.data.frame(nlme::Machines); ma$Worker <- factor(as.character(ma$Worker));
↪  ma$Machine <- factor(as.character(ma$Machine))
round(tapply(ma$score, list(Worker = ma$Worker, Machine = ma$Machine), mean), 1)
```

```
#>       Machine
#> Worker    A    B    C
#>      1 52.6 62.9 67.2
#>      2 52.6 59.6 61.8
#>      3 59.5 68.0 70.8
#>      4 51.2 62.7 64.8
#>      5 51.4 65.1 71.7
#>      6 46.8 43.6 61.3
```

(a) Justify fixed and random effects for `Machine` and `Worker`. Fit (REML) `score ~ Machine + (1 | Worker)` and `score ~ Machine + (1 | Worker) + (1 | Worker:Machine)` and write both models. Interpret the variance components of the second model (what does σ^2_{WM} mean?).

(b) Test the worker-by-machine variance $\sigma^2_{WM} = 0$ (REML fits with the same fixed effects), giving T, the reference distribution and the p-value; compare AIC.

(c) Test the machine effect (3 levels, 2 df) with the correct likelihood, and compute the (invalid) statistic from the two REML fits for comparison. Then repeat the test with an F test with Satterthwaite or Kenward–Roger degrees of freedom (`lmerTest`, `pbkrtest`), and compare with the exact F test of the balanced split-plot ANOVA, `aov(score ~ Machine + Error(Worker/Machine))`. Why is the LRT p-value too small with only 6 workers?

(d) Inspect the table of cell means: worker 6 on machine B behaves differently. Refit the model without that cell. How do the variance components change, and what does that tell you about the interaction variance? What would you do in a real analysis?

(e) Predict, with a 95% prediction interval, the score of a **new worker** on machine C, using $\text{Var} = x_0^\top \text{Var}(\hat\beta) x_0 + \sigma^2_W + \sigma^2_{WM} + \sigma^2$, and compare with the point prediction for **worker 3** on machine C.

(f) *Communicate.* In about 120 words, tell the factory manager whether machine C is better than A and B, whether this holds for every worker, and how sure you are.

7 A logistic mixed model: verbal aggression

The data `lme4::VerbAgg` contain the answers of 316 people (243 women, 73 men) to 24 items describing situations in which one might react verbally with aggression (`r2`: Y/N response). Items vary in the behaviour type (`btype`: curse, scold, shout) and in whether the person is the cause of the frustration (`situ`: other, self).

```
data(VerbAgg, package = "lme4")
fit_va <- glmer(r2 ~ Gender + btype + situ + (1 | id), family = binomial, data =
↪  VerbAgg)
summary(fit_va)$coefficients
```

```
#>              Estimate Std. Error  z value    Pr(>|z|)
#> (Intercept)  1.30732   0.103665  12.6110  1.8357e-36
#> GenderM      0.29522   0.186172   1.5858  1.1280e-01
#> btypescold  -1.03112   0.068098 -15.1416  8.6092e-52
#> btypeshout  -1.99567   0.073354 -27.2059 5.5243e-163
```

```
#> situself    -1.00462   0.056959 -17.6378  1.2634e-69
```

```
VarCorr(fit_va)
```

```
#>  Groups Name        Std.Dev.
#>  id     (Intercept) 1.3
```

(a) Write the model. Interpret `btypeshout`, `situself` and $\hat{\sigma}_b$ on the odds scale; give a 95% interval for the multiplicative effect on the odds across people implied by $\hat{\sigma}_b$ (the range within which 95% of people's baseline odds lie relative to the average person).

(b) Compute the latent-scale ICC. Is this the correlation between two binary answers of the same person? Why (not)?

(c) Refit with `nAGQ = 10` and compare the estimates with Laplace. Is the Laplace approximation adequate here? Explain with respect to the data structure.

(d) The coefficient of `GenderM` is 0.295. Interpret it as an odds ratio with a 95% Wald interval and test it with an LRT. A newspaper writes "men are 34% more likely to respond aggressively". Critique the statement.

(e) **Conditional vs marginal.** Fit `glm(r2 ~ Gender + btype + situ, family = binomial)` and compare its coefficients with the GLMM coefficients and with the approximation $\hat{\beta}_C/\sqrt{1+0.346\hat{\sigma}_b^2}$. For a woman, the item type "curse", situation "other": compute the probability of answering Y for a *typical* person ($b=0$) and the *population-average* probability (numerical integration).

8 Laplace and quadrature for one cluster of counts

A cluster has three counts $y=(3,1,2)$ and the model is $y_j \mid b \sim \text{Poisson}(e^b)$, $j=1,2,3$, $b\sim N(0,1)$ (no covariates).

(a) Write the likelihood $L=\int \prod_j f(y_j \mid b)\,\phi(b)\,db$ and $h(b)=\log\{\prod_j f(y_j \mid b)\phi(b)\}$. Give $h'(b)$ and $-h''(b)$.

(b) Find the mode $\hat{b}$ by Newton's method (start at 0.5, three iterations) and compute $H=-h''(\hat{b})$ and the Laplace approximation of L.

(c) Compute L numerically (`integrate`) and the relative error of Laplace. Implement the adaptive Gauss–Hermite rule with $K=1,3,5,9$ nodes (nodes and weights from the eigenvalues and eigenvectors of the Jacobi matrix with off-diagonals $\sqrt{k/2}$, as in the class notes) and report the relative errors.

(d) **Derivation.** Show that, for a canonical link, the Laplace log-likelihood contribution of a cluster with $q=1$ random effect can be written $\sum_j \log f(y_j \mid \hat{b}) - \hat{b}^2/(2\tau^2) - \frac{1}{2}\log\{1+\tau^2\sum_j \hat{\mu}_j\}$ for $b\sim N(0,\tau^2)$ and Poisson f, and verify it numerically for the cluster above. (Here h contains the term $-\frac{1}{2}\log 2\pi\tau^2$ from the prior.)

(e) Without computing: how would the relative error of the Laplace approximation change if the cluster had $(30,10,20)$ instead (same ratios, ten times larger counts)? Why?